

ZAALI Mohamed

AI Software Engineer

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EDUCATION

ENSAM Rabat - State Engineer Student in Data Science, AI & Digital Health

Sep 2023–Present

PROFESSIONAL EXPERIENCE

Syntax - Casablanca - AI Agent Developer Intern

Feb 2026 - Present

Building an extensible MCP server connecting AI agents to Microsoft 365 via Microsoft Graph API.

- Reduced agent configuration complexity by **architecting a preset system** that ships context-aware tool sets.
- Built a **5-step pipeline** that auto-generates typed MCP tools from Microsoft's OpenAPI spec.
- Enabled any M365 service to be onboarded with **zero code changes** by designing a JSON-driven extensibility architecture.

Al Barid Bank - Rabat - Full Stack Developer Intern

Jul 2025 - Sep 2025

Design and development of a platform for analyzing POS, SoftPOS, and e-commerce transactions.

- Accelerated transaction data access by building a **secure REST API** serving aggregated POS, SoftPOS, and e-commerce metrics.
- Empowered analysts to explore trends by delivering a **responsive dashboard** with interactive charts and drill-down navigation.
- Ensured long-term maintainability by designing the full **UML software architecture** before implementation, reducing rework risk.

Ministry of Youth, Culture and Communication - Rabat - Data Science Intern

Jul 2024 - Sep 2024

Development of a predictive analytics system for cultural heritage preservation.

- Enabled proactive preservation decisions by building a **predictive model** forecasting cultural heritage degradation risk.
- Made insights accessible to non-technical experts by integrating model predictions into an interactive **Flask web application**.
- Ensured production availability by deploying the application on an **Ubuntu server** with Unicorn and Nginx.

PROJECTS & ACCOMPLISHMENTS

- Project Architect Agent - Autonomous Project Documentation Agent:** Agentic system that takes a project idea, runs a **research loop** per implementation step, and generates a structured **Notion workspace**. | Technologies: `LangGraph` `OpenAI API` `Gradio` `Notion API`
- AI Invoice Processing Agent - Automated Invoice Approval Workflow:** Agent that extracts, validates, and approves PDF invoices with business rule enforcement and email notifications. | Technologies: `LangGraph` `Gradio` `OpenAI API` `Gmail API`
- Twitter MCP Server - Tool for AI Agent Social Media Automation:** Exposes Twitter/X tools (post, thread, search) to any MCP-compliant AI agent via Streamable HTTP and SSE endpoints. | Technologies: `FastMCP` `Python` `Twitter API`
- El-Qess Restaurant AI Assistant - Langflow-Powered Restaurant Agent:** Conversational agent handling **FAQ via RAG**, table reservations, and confirmation emails. | Technologies: `LangFlow` `OpenAI API` `Chroma DB` `Google Calendar MCP` `Gmail MCP`
- AL-MUQARRIR - Automatic Transcription & Meeting Minutes Generation:** Web platform for **real-time transcription**, speaker **diarization**, and automatic generation of **meeting minutes** with **semantic analysis**. | Technologies: `Whisper` `PyAnnote` `Gemini` `Flask` `Angular`
- Cura - Disease & Medication Tracking:** Web application for treatment management with AI-powered **drug interaction detection** and generation of patient-friendly alerts. | Technologies: `PyTorch` `Gemini API` `Flask` `SQLAlchemy` `Bootstrap` `SQLite`
- LipReader - Lip-Reading Application:** Application converting lip movements into text. | Technologies: `TensorFlow` `Streamlit`
- Real-Time Network Intrusion Detection System:** System capturing and analyzing network packets in real-time. | Technologies: `Python` `Kafka` `PyShark` `Pandas` `Scikit-learn` `Joblib`

TECHNICAL SKILLS

AI Engineering: `LangGraph` `LangChain` `LangFlow` `FastMCP` `OpenAI API` `Gemini API` `RAG`

Programming Languages: `Python` `Java` `TypeScript` `JavaScript`

ML/DL: `PyTorch` `TensorFlow` `Scikit-learn` `NLP`

Databases: `PostgreSQL` `MySQL` `MongoDB` `SQLite`

Web Tech & Frameworks: `HTML` `CSS` `Bootstrap` `Angular` `Flask` `FastAPI` `Spring Boot` `Streamlit` `Gradio`

EXTRACURRICULAR ACTIVITIES

Learning Experience Architect - Fatal Error Club ENSAM Rabat

SEP 2024 - JUN 2025

- Led workshops in data science, machine learning, and software development; co-created innovative programs.
- Optimized training outcomes through continuous evaluation and refinement.

CERTIFICATIONS

• CCNA: Introduction to Networks

• Data Structures & Algorithms in Python

• Python Programming Fundamentals

• SQL Fundamentals

• Introduction to Data

LANGUAGES

• **English:** Fluent

• **French:** Fluent

• **Arabic:** Native